The Coherence Burden Principle: Support, Lifetime, and Structural Admissibility Across Domains

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Abstract

Darke Theory has developed as a layered structural framework in which recursive differentiation, closure, and admissibility determine which configurations can be stably realised and which cannot. Across the corpus, a recurring logic appears in both mathematical exclusion papers and empirical precursor studies: persistent pathological regimes require organised coherence, transport, or structural support whose maintenance is constrained by finite closure-bearing capacity. At the same time, several empirical domains show that coherence may arise transiently as a measurable precursor to reconfiguration without remaining persistently supportable. This paper formalises that recurring architecture as the *Coherence Burden Principle*. We define coherence, support, coherence lifetime, coherence burden, and structural admissibility in a domain-transportable way, and distinguish three regime classes: inadmissible persistent regimes, admissible transient precursor regimes, and structurally informative residual regimes. We then state a general theorem-schema: whenever a candidate regime requires coherence whose support demand exceeds the admissible lifetime or capacity of the governing system, the regime is structurally inadmissible; when coherence is real but only finitely supportable, it may appear only as a transient precursor or local restructuring state. The aim is not to replace domain proofs, but to extract the general law already implicit in the corpus and provide a unified bridge between foundational ontology, structural mathematics, cross-domain exclusion, empirical precursor evidence, and coherence-sensitive observability.

1 Introduction

Across the Darke Theory corpus, one recurring structural pattern has become increasingly difficult to ignore. In mathematical exclusion papers, pathological regimes are not ruled out by brute-force exhaustion, nor by appeal to impossibility claims imposed from outside the domain. Rather, they are shown to depend upon forms of coherence, support, or organised persistence whose sustaining burden exceeds what the governing system can admissibly carry. In empirical precursor papers, by contrast, coherence is not excluded. It appears as a real and measurable organisation before rupture, reconfiguration, failure, or release, but only within a finite support horizon. In coherence-sensitive sensing work, still another variation appears: meaningful structure survives in observables that remain responsive to organisation even after ordinary second-order summaries have degraded into noise, clutter, or collapse.

These are not superficial similarities. They point to a common regime logic already present across the corpus but not yet isolated in its own right. That logic is simple to state, but far-reaching in consequence: persistence requires coherence; coherence requires support; support is constrained; and admissibility depends upon whether that burden can in fact be borne for the lifetime required by the regime in question. Once this is seen clearly, a large fraction of the later corpus becomes legible as a sequence of domain-specific instantiations of one deeper structural principle.

The present paper extracts that principle explicitly. It does not replace the foundational sequence, the mathematical exclusion papers, or the empirical precursor studies. Nor does it

flatten multiple technical domains into one artificial formal language. Its aim is narrower and more structural. It identifies the recurring variables by which Darke Theory has already been operating, often implicitly: support, lifetime, burden, and admissibility. What earlier papers have shown in specific settings is here drawn into a single transportable framework.

This paper is necessary for at least four reasons. First, the corpus has already moved beyond the stage at which admissibility can remain a purely general slogan. The later papers do not merely say that certain regimes are inadmissible. They repeatedly show *why* they are inadmissible: because persistence would require organised burden that cannot remain supportable across the necessary scale, depth, or duration. Without naming this mechanism directly, the corpus risks appearing to consist of many parallel applications rather than one increasingly unified structural programme.

Secondly, the corpus now contains two seemingly different branches that should no longer be left only loosely connected. The first is the exclusion branch, where certain regimes are argued to be structurally impossible as persistent branches. The second is the empirical precursor branch, where coherence is directly observed before transitions, failures, or instabilities. Without a mediating principle, these can appear to pull in opposite directions: one seems to say that persistent coherence burdens fail, while the other seems to say that coherence is real and recurrent. In fact they are compatible. The distinction is not between real and unreal coherence. It is between coherence that can be borne persistently, coherence that can be borne only transiently, and coherence that remains visible only in the correct observational channel.

Thirdly, a transportable burden-language makes the corpus more reviewable. If a future paper claims that a candidate regime is inadmissible, a serious reader should be able to ask: what coherence is required, what support carries it, what lifetime is necessary, what constraints degrade it, and why is the burden greater than the admissible budget. Likewise, if a paper claims to detect precursor structure, the proper questions are: what kind of coherence is being measured, over what horizon does it remain organised, and why is it interpreted as a transient branch rather than stable background? A general paper of this kind therefore sharpens the hostile-review interface of the entire corpus.

Fourthly, this paper solves an internal problem of lineage. Darke Theory has already built a foundational path from recursive differentiation through closure into admissibility and stable structure. But the corpus has not yet fully articulated how later domain papers sit downstream of that sequence. The current paper supplies that missing bridge. It argues that support and lifetime are the operational variables by which closure and admissibility become visible at the regime level. Put differently, if closure is what allows a structure to stabilise, and admissibility is what determines whether such stabilisation is genuinely supportable, then coherence burden is the practical currency in which that question is paid.

The paper therefore advances a regime-level claim rather than a domain-specific theorem. It argues that structural admissibility depends not merely on whether a regime can be described, locally approximated, or transiently instantiated, but on whether the coherence required for its persistence can be supported for the necessary lifetime under the closure, refinement, and constraint structure of the governing system.

From this standpoint, the paper has five linked goals.

1. To state a general principle linking coherence, support, lifetime, and admissibility.

2. To distinguish persistent pathological regimes from transient precursor regimes and structurally informative residual regimes.

3. To supply a general theorem-schema capable of organising both exclusion results and precursor observations.

4. To show how a broad range of existing Darke Theory papers can be read as instantiations of the same structural law.

5. To provide a disciplined review framework so that future burden-based claims can be tested at their true structural hinge rather than only at the level of rhetoric.

It is equally important to state what this paper does not attempt. It is not a universal explanatory programme. It does not claim that every mathematical, physical, or empirical problem can be translated meaningfully into admissibility language. It does not generate dynamical laws, fix constants of nature, or replace successful domain theories within their native scope. It does not convert empirical suggestiveness into proof. It applies only where coherence, persistence, closure, and contradiction-inducing forbidden regimes can be stated with sufficient clarity for burden and support to be more than metaphorical terms. Its selectivity is therefore part of its discipline, not a weakness to be hidden.

The structure of the paper is as follows. Section 2 places the current argument downstream of the foundational Darke Theory sequence and explains why burden is a natural consequence of refinement, closure, and admissibility. Section 3 defines the operational vocabulary of coherence, support, lifetime, burden, and finite admissibility budget. Section 4 states the Coherence Burden Principle itself and explains why it functions as a regime filter rather than a local yes-or-no test. Section 5 introduces the three regime classes implicit in the current corpus. Section 6 gives the main transportable theorem-schema. Section 7 expands the internal logic of support, lifetime, and budget. Section 8 relates that schema to the exclusion branch, the precursor branch, and the observability branch. The paper closes with hostile-review criteria, explicit scope limits, and a concluding statement of the corpus-level significance of the principle.

2 Foundational Setting

The present paper stands downstream of the foundational Darke Theory sequence in which recursive differentiation generates structured distinctions, closure stabilises selected patterns, and admissibility determines which branches can be genuinely supported rather than merely described. In that lineage, persistent structure is not primitive. It is earned. Objecthood, stable observability, regime-level identity, and durable structure all appear only after the relevant burden has been carried successfully through admissible refinement.

Compressed to its structural minimum, the foundational lineage may be written as

$$\text{recursive differentiation} \implies \text{closure} \implies \text{admissibility} \implies \text{persistent structure and regime selection.}$$

This compressed sequence is not a substitute for the foundational papers. It is only the load-bearing path required for the present argument. The key point is that nothing in this path makes persistence free. Recursive differentiation proliferates distinctions and therefore the possibility of structure, but not yet stable structure. Closure provides a mechanism by which some patterns stabilise rather than disperse, but closure itself does not guarantee unlimited persistence. Admissibility then enters not as a decorative final label but as the decisive question: can this branch be borne without contradiction, fragmentation, or structurally unsupported burden?

The present paper extracts one consequence that becomes increasingly visible in later corpus work: every stable regime carries an implicit support problem. If a candidate branch is to persist, some organisational resource must hold it together over the relevant scale, load, or duration. In some domains this resource appears as phase alignment. In others it appears as scale-local transport organisation, large-scale gauge coherence, semantic reuse, closure-bearing integration, or estimator-sensitive residual structure. The specific technical form changes. The structural role does not.

This is why the notion of burden belongs downstream of the foundational papers rather than being an external invention. Once recursive differentiation is understood to generate branching structural possibilities, closure is understood to stabilise some of them, and admissibility is

understood to filter which stabilisations are genuinely supportable, the next natural question becomes unavoidable: what does it *cost*, structurally, for a regime to persist? The later corpus repeatedly answers that question in domain-specific ways. The current paper names and organises that answer.

2.1 From closure to carrying capacity

Closure stabilises patterns, but only where stabilisation is structurally supportable. That supportability is not binary in a crude sense. It depends on the carrying capacity of the system. Some regimes demand little: local, weak, or short-lived organisation may be easily sustained. Others demand much more: long-range phase coordination, repeated cross-scale transport alignment, persistent holonomy organisation, or robust semantic reuse across instance variability. Once this is recognised, admissibility ceases to look like an abstract yes-or-no judgement and begins to look like a question of carrying capacity.

The phrase *carrying capacity* is deliberately structural rather than mechanistic. It does not refer only to energy, only to width, or only to combinatorial size. In one domain, carrying capacity may be the ability to maintain aligned transport without fragmentation. In another, it may be the ability to preserve coherence against non-commutative drift. In another, it may be the stability of semantic reuse under adversarial variation. What matters is not the uniformity of mechanism, but the repeated appearance of a bounded structural ceiling beyond which persistence can no longer be supported.

2.2 Realisability versus formal descriptibility

A further foundational consequence is that formal descriptibility and structural realisability must not be confused. A regime may be expressible in equations, combinatorial descriptions, or analytic continuation. It may even be locally approximable or briefly observable. None of this yet shows that it can exist as a stable branch. The foundational DT distinction between what can be formally written and what can be structurally borne is precisely what later burden arguments exploit.

This distinction becomes especially important when reading the proof layer of the corpus. In those papers, the target is often not a lack of local construction but a failure of persistent realisability. The candidate regime is shown to require support whose burden cannot remain admissible under the system's own structural constraints. The present paper does not introduce a new criterion from nowhere. It draws out the operational meaning of a distinction that has already been foundational from the beginning.

2.3 Why lifetime must be added

Support alone is not enough to describe the regime problem. A system may support a given form of coherence briefly and yet fail to support it persistently. That is why lifetime must be introduced as a first-class variable. Closure is not merely about whether a pattern can appear, but whether it can survive through the relevant refinement depth, scale span, or temporal interval. Some later corpus papers already exhibit this implicitly by showing that a coherence requirement must be renewed indefinitely or across many layers, while the support available to it decays, fragments, or remains summably bounded.

Lifetime therefore measures the persistence horizon relevant to admissibility. It is not a mere duration parameter. In different domains it may mean elapsed time, cascade depth, resolution scale, geometric distance, compositional depth, or reuse horizon. What unites these cases is that persistence is meaningful only relative to some horizon, and burden is meaningful only relative to what must be sustained across that horizon.

2.4 The downstream foundational claim

The downstream claim of the present paper can now be stated plainly. If recursive differentiation generates candidate structures, closure stabilises some of them, and admissibility filters which stabilisations are genuinely supportable, then a large class of later arguments will naturally take the following form:

> A candidate regime requires organised coherence for persistence; the required coherence induces a burden over some lifetime; the system provides only finite, unstable, or summably bounded support; therefore persistent realisation of the regime is structurally inadmissible, or else possible only as a transient local manifestation.

That pattern is not an optional interpretive overlay. It is what makes the later exclusion papers and precursor papers look like descendants of the same foundational theory rather than disconnected applications.

3 Core Definitions

The present section introduces the operational vocabulary of the paper. These definitions are intentionally domain-transportable. They are not meant to erase technical differences between analytic number theory, gauge theory, nonlinear transport, empirical precursor studies, or signal processing. They are meant to identify the structural roles that recur across those domains.

Definition 1 (Coherence)**.** *Coherence is the persistence of compatible structure across the degrees of freedom relevant to a candidate regime. It is the organisational alignment by which persistence is paid for. Depending on domain, coherence may take the form of phase alignment, transport stability, gauge organisation, semantic reuse, or closure-bearing relational support.*

Coherence should not be confused with mere regularity. A signal may look smooth without carrying the organised structure required for persistence, and a system may be highly irregular while still exhibiting exactly the kind of aligned burden relevant to a given regime. The point of the definition is therefore not aesthetic order but support-bearing organisation. In the Riemann setting, coherence appears through structured prime-phase alignment. In precursor work on EEG or self-organised criticality, coherence appears as phase organisation that outstrips null expectations. In transport problems, coherence may appear as sustained organisation of mode interaction or scale-local transfer. These are not the same mathematical object. They are the same structural currency.

Definition 2 (Support)**.** *Support is the effective structural capacity of a system to sustain coherence without loss of admissibility. It includes whatever bounded resources the domain makes relevant: integrating width, symmetry allowance, transport stability, closure-bearing capacity, finite variance, semantic compression budget, or analogous sustaining structure.*

Support must be understood as carrying capacity rather than simple enabling presence. It is not enough that a system contains some mechanism compatible with coherence. The relevant question is whether it can carry the burden induced by persistence across the horizon that matters. A domain may support local organisation while failing to support global organisation, or support brief alignment while failing to sustain long-run renewal. Support is therefore always relative to burden. There is no meaningful support in the abstract.

Definition 3 (Coherence lifetime)**.** *The coherence lifetime of a regime is the maximal interval, scale-depth, or combinatorial extent over which the coherence required for that regime can remain supportable before degradation, fragmentation, or constraint-induced collapse renders it inadmissible.*

Lifetime is the persistence horizon of coherence. It is the answer to the question: how long, or how far, can the relevant organisation be borne before it ceases to remain structurally supportable? In one domain this may be measured temporally. In another it may be measured in scale cascade, iteration depth, or analytic extent. What matters is not the units but the horizon of persistence. A regime that requires indefinite support is exposed to a different admissibility test from a regime that requires only a short burst of organisation.

Definition 4 (Coherence burden)**.** *The coherence burden of a candidate regime is the total support demand required to maintain the regime across its relevant lifetime, scale-span, or persistence depth. A regime carries high coherence burden when its persistence requires unusually aligned, organised, or repeatedly renewed support rather than merely local or transient structure.*

Burden is the central transportable variable of the present paper. It measures not simply whether coherence exists, but what must be *paid* to keep it going. Burden grows when organisation must be renewed across many layers, kept aligned against fragmentation, or maintained despite hostile background dynamics. Burden is therefore naturally sensitive to scale, duration, coordination width, and renewal cost. It is what later exclusion arguments keep rediscovering in different language.

Definition 5 (Structural admissibility)**.** *A regime is structurally admissible if it can be stably realised within the closure, persistence, and refinement constraints of the governing system without requiring structurally unsupported burden.*

This definition preserves the foundational distinction between formal descriptibility and genuine supportability. A regime may be expressible and even locally approximable while remaining inadmissible as a stable branch. Structural admissibility is therefore not a question about symbolic representation alone. It is a question about whether the system can actually bear the burden induced by the regime's persistence requirements.

Definition 6 (Finite admissibility budget)**.** *A system has a finite admissibility budget for a regime when the support available to sustain the relevant coherence is not arbitrarily extensible, but bounded, unstable, summable, or otherwise limited across scales, distances, or combinatorial layers.*

The language of budget is especially useful because it prevents admissibility from being read as a purely local obstruction. A budget is a system-level carrying ceiling. It may be finite because support decays with scale, because integrative width is bounded, because nonlinearity disperses coordination, because semantic reuse is fragile, or because closure cannot be maintained indefinitely. The exact mechanism varies. The structural consequence does not: a regime cannot remain admissible once its support demand exceeds the total budget that the system can provide.

Definition 7 (Constraint)**.** *Constraint is the family of structural limitations that erode, bound, fragment, or cap the support available to a regime. If coherence names the support required by persistence, constraint names the mechanism of its erosion.*

Constraint is not a negative afterthought. It is the partner concept without which burden cannot be understood. Every burden argument contains an account, whether explicit or implicit, of how support is degraded. In some settings constraint appears as cancellation or summable depletion. In others it appears as dispersive transport, non-commutative drift, noise, locality, bounded width, or adversarial instability. Constraint is the reason a system with nonzero support need not have unlimited support.

4 The Coherence Burden Principle

The preceding definitions allow the central law to be stated in direct form.

Principle 8 (The Coherence Burden Principle)**.** *A candidate regime is structurally admissible only if the coherence required for its persistence can be supported for the necessary lifetime within the closure, refinement, and constraint structure of the governing system.*

This principle is a sharpening, not a replacement, of the general admissibility criterion. Its central claim is that admissibility is not decided solely by local constructibility, nor solely by formal definability, nor solely by transient observability. It is decided at the level of sustained burden. A regime must be judged not only by whether it can appear, but by whether the support required to keep it going can actually be carried across the horizon that gives the regime its claimed significance.

4.1 Why coherence is the currency of persistence

Persistence is never free. If a pattern is to remain stable rather than dissolve into local fluctuation, cancellation, fragmentation, or generic drift, some aligning structure must hold it together. That structure is what this paper names coherence. The term is chosen not to force identical mathematics upon different domains, but to isolate a shared structural role. In every case, coherence names the organisational alignment that persistence demands.

This is why coherence is best understood as the currency of persistence. A regime can persist only by spending some form of coordinated support. The more exceptional the regime, the more exacting this expenditure tends to become. Local coherence may be cheap. Persistent cross-scale or cross-instance coherence is not. Once a paper identifies a forbidden regime and derives the structured support required for its persistence, it has effectively translated the problem into the currency of burden.

4.2 Why support is never effectively infinite

The second point of the principle is that support must be treated as bounded unless a domain-specific reason is given otherwise. Darke Theory does not begin from a fantasy of unlimited carrying capacity. It begins from refinement, closure, and admissibility, which together imply that stabilisation is selective and burden-sensitive. Support may be rich, but it is not presumed unbounded. Across the corpus, support limits appear in many forms: bounded variance, finite width, decaying transport organisation, instability of large-scale coordination, summability across scales, fragility of semantic reuse, and the collapse of order under adversarial or nonlinear conditions.

This boundedness is what makes admissibility a genuine filter. If support were effectively infinite, any coherence burden could in principle be carried, and the distinction between formal possibility and structural realisability would erode. The repeated appearance of finite or degradable support is therefore not an empirical accident in the later corpus. It is the practical face of admissibility itself.

4.3 Why lifetime matters as much as burden

A regime may require only modest support at any single local step and yet remain inadmissible because the support must be renewed indefinitely, or across too many scales, or across too great a distance, or across too many reuse instances. Burden is therefore inseparable from lifetime. A support demand must always be asked *for how long*, or *across what depth*, if its admissibility is to be judged properly.

This is one reason the paper insists on coherence lifetime as a first-class variable. Without lifetime, support can be misread as a local property, and transient coherence can be mistaken either for proof of stable admissibility or for a counterexample to burden logic. In fact, lifetime is what separates the persistent pathological branch from the transient precursor branch. A

system may support a burst of organisation without supporting the indefinite renewal required for stable persistence.

4.4 Admissibility as regime filter

The principle also clarifies how admissibility should be understood. Admissibility is not a local yes-or-no property attached to an isolated event. It is a regime filter. It asks whether the system can support the persistence conditions necessary for a branch to count as genuine. This makes admissibility a higher-level structural criterion. It is evaluated not at the level of raw possibility, but at the level of sustained and closure-compatible realisability.

This helps explain why the same principle can organise both exclusion and precursor work. In the exclusion case, a regime fails because its required burden outstrips admissible support over the relevant lifetime. In the precursor case, a regime appears because local or short-run support is available, but it does not become a stable branch because the necessary lifetime cannot be carried. The same regime filter is operating in both cases.

4.5 Transient coherence is not a counterexample

A common misunderstanding would be to treat empirical precursor coherence as a refutation of burden-based inadmissibility. The present framework rejects that reading. Transient coherence is not a counterexample to the principle. It is one of its positive manifestations.

If the system can carry coherence only briefly, then the correct prediction is not that coherence never appears, but that it appears as a finite-lived, localised, or transitional organisation rather than as a permanently supportable branch. This is why the precursor papers belong naturally inside the same structural architecture as the exclusion papers. They show the intermediate regime in which support is real but not indefinitely renewable.

4.6 Residual observability as a third manifestation

There is also a third branch that the principle helps isolate. In some settings the key issue is neither whether a regime is persistently admissible nor whether it appears as a precursor. Instead, the issue is whether structure remains detectable at all after ordinary summary statistics have degraded. Here the relevant insight is that coherence-sensitive observables may preserve meaningful organisation even when energy-only, amplitude-only, or second-order summaries have become uninformative.

This branch is important because it shows that coherence is not only a burden variable. It is also an observability variable. The very organisational structure whose persistent burden may exceed admissible support can nevertheless remain the most informative channel through which a system is understood. This is one reason the current paper aims at a regime theory rather than a merely exclusionary framework.

4.7 Operational reading of the principle

Taken operationally, the Coherence Burden Principle instructs future work to proceed in a disciplined order.

1. Identify the candidate regime whose status is in question.
2. Determine what coherence that regime requires for persistence, not merely for local appearance.
3. Determine what support resources must carry that coherence.
4. Determine the lifetime or depth over which those resources must remain available.

5. Determine how the governing constraints of the system degrade, bound, or fragment that support.

6. Compare burden to admissible budget and classify the regime accordingly.

This sequence is the methodological heart of the paper. It converts the principle from a philosophical statement into a transportable review framework.

4.8 Summary

The Coherence Burden Principle may therefore be read as the regime-level operational form of a longer foundational lineage. Recursive differentiation creates structural possibility. Closure stabilises selected patterns. Admissibility filters what can truly be borne. The present paper adds that the decisive variable at the regime level is the burden induced by coherence across a required lifetime under finite support. Once this is stated plainly, the later Darke Theory corpus becomes much more unified: what differs from paper to paper is the technical form of coherence, support, and constraint, not the basic logic by which persistent branches, transient precursors, and residual observables are classified.

5 Regime Classification

The proof layer of the corpus strongly suggests that candidate regimes do not all stand in the same relation to admissibility. Some are structurally impossible as persistent branches. Others are real and measurable, but only as finite-lived precursor states. Still others matter chiefly because they preserve information in observables after conventional summaries fail. To prevent these cases from being conflated, the present paper introduces a three-part regime classification.

Definition 9 (Persistent pathological regime)**.** *A persistent pathological regime is a branch whose continued existence would obstruct a target theorem, stable law, or admissible large-scale configuration, and whose maintenance requires coherence across sufficient lifetime or scale to become support-limited.*

Definition 10 (Transient precursor regime)**.** *A transient precursor regime is a localised, finite-lived organisation of coherence that appears before transition, rupture, or reconfiguration, but is not sustained as a stable long-run branch.*

Definition 11 (Structurally informative residual regime)**.** *A structurally informative residual regime is one in which coherence-sensitive observables retain meaningful organisation even when conventional amplitude-, energy-, or second-order summaries collapse or become uninformative.*

These three classes do not exhaust every possible system behaviour, but they organise a large portion of the present corpus and clarify how apparently different papers can sit within one regime architecture.

5.1 Persistent pathological regimes

In a persistent pathological regime, the central issue is not whether a pattern can occur locally, nor whether it can be written down formally, but whether it can persist in the strong sense required to obstruct the target result. Such regimes are typically load-bearing in exclusion papers. Off-line zeros in the Riemann setting, massless large-scale coherence in Yang–Mills, singular transport persistence in Navier–Stokes, or robust semantic compression across adversarial instance families in the complexity setting all belong to this class.

The reason this class matters is that it forces burden to the foreground. A local event may be cheap. A persistent pathological branch is not. It must remain organised across the full

horizon relevant to the theorem or law it threatens. The hostile-review question for this class is therefore sharp: does the forbidden regime really require the kind of coherence claimed, and if so, can the system actually support it across the required horizon?

5.2 Transient precursor regimes

A transient precursor regime is not a failed pathology. It is a positive structural class in its own right. Here coherence is real, measurable, and often diagnostically useful, but only within a finite window before a major reconfiguration. Once the event occurs, the coherence weakens, collapses, or redistributes. This class is particularly important for the empirical branch of the corpus, because it prevents a false dichotomy between "coherence is impossible" and "coherence is stable". The correct alternative is often that coherence is real but finite-lived.

The hostile-review question for this class is correspondingly different. One must ask whether the detected organisation is genuinely coherence-sensitive rather than a trivial reflection of power or amplitude, and whether it is localised in the way the theory expects rather than persisting as a generic background feature.

5.3 Structurally informative residual regimes

The third class becomes visible when the problem is not whether a regime persists, but whether structure can still be *seen* at all once ordinary summaries have smeared it out. In such cases coherence is important not because it defines a forbidden or precursor branch, but because it survives in the correct observational channel when other channels fail. This class is especially relevant to signal-rich, cluttered, or measurement-limited systems. It shows that coherence is both a burden variable and an observability variable.

The hostile-review question here is: does the chosen observable genuinely preserve admissibility-relevant structure, or is it merely another summary statistic with a more complicated form?

5.4 Why the classification matters

Without this classification, the corpus risks two forms of confusion. First, exclusion papers may be read as though they deny the reality of all coherence, when in fact they deny only the persistent admissibility of certain high-burden branches. Secondly, precursor papers may be misread as counterexamples to inadmissibility, when in fact they typically show finite-lived support rather than indefinitely renewable support. The three-part classification resolves both confusions.

It also creates a better internal map of the corpus. The later work is no longer best described as "foundations plus applications". It is better described as a regime programme with at least three branches:

- an exclusion branch, in which persistent pathology is ruled out by excessive coherence burden;
- a precursor branch, in which coherence appears as a finite-lived transition signal;
- an observability branch, in which coherence-sensitive measurements preserve structure that conventional summaries discard.

6 General Theorem-Schema

The present section states the main transportable kernel of the paper. The aim is not to replace domain mathematics with one grand formalism, but to make explicit the shared logical structure that multiple domain papers already instantiate.

Theorem 12 (Finite Support Exclusion Schema)**.** *Let R be a candidate persistent regime in a system S. Suppose:*

1. *R requires a definable form of coherence C_R for its persistence;*
2. *the persistence of R induces a coherence burden B_R over its relevant lifetime or scale-span;*
3. *the governing structure of S supplies at most an admissible support capacity A_S, with either bounded total support or summably limited support across the relevant persistence depth;*
4. *$B_R > A_S$ in the sense made precise by the domain.*

Then R is structurally inadmissible in S.

Interpretive proof sketch. The argument is indirect. One first identifies the obstructive regime R, not as a bare event, but as a pattern that must be sustained. One then derives the form of coherence required for that sustaining. The next step is to show that the system does not grant arbitrary support for such coherence: the relevant support is bounded, summably limited, unstable, or otherwise constrained. If the sustaining burden exceeds what the governing structure can bear, then persistence of R cannot be housed stably within the system. Hence R is inadmissible. □

This is a theorem-schema rather than a single theorem in the narrow technical sense. Its role is not to replace domain mathematics, but to state the exact general structure that domain-specific arguments must instantiate if they are to count as genuine coherence-burden exclusions.

6.1 Schematic notation

To make the schema more explicit, one may regard the coherence burden B_R as an abstract burden functional depending on the coherence form, the persistence horizon, and the renewal cost required by the regime:

$$B_R = \mathcal{B}(C_R, L_R, \rho_R),$$

where L_R denotes the relevant lifetime or depth horizon, and ρ_R denotes the renewal or maintenance demand required to keep the coherence in place across that horizon. Likewise, one may regard the admissible support capacity as an abstract support functional

$$A_S = \mathcal{A}(S, \Gamma_S),$$

where Γ_S denotes the governing constraint structure of the system. The schema then compares burden to support not numerically in a universal sense, but structurally: the burden induced by persistence must not outstrip the support admissibly available.

This notation is schematic on purpose. Its value lies not in numerical universality but in exposing what later domain papers must make precise if they are to instantiate the schema rigorously.

Proposition 13 (Bounded-support obstruction)**.** *If the support functional $\mathcal{A}(S, \Gamma_S)$ is bounded while the burden functional $\mathcal{B}(C_R, L_R, \rho_R)$ grows without admissible compensation across the relevant persistence horizon, then R cannot be a persistent admissible regime.*

Proof. If $\mathcal{A}$ is bounded and $\mathcal{B}$ exceeds it over the persistence horizon relevant to R, then the coherence required for R cannot be stably maintained. Hence persistence of R fails at the level required for regime admissibility. □

Proposition 14 (Finite-lifetime obstruction)**.** *If the coherence required by R is supportable only over a finite horizon L^*, but persistence of R requires support beyond L^*, then R is inadmissible as a persistent regime, even if local realisations occur below that horizon.*

Proof. The regime significance of R depends on persistence beyond the finite horizon of supportability. If support is exhausted or degraded beyond L^*, then the regime cannot be borne in the strong sense required for persistence. Hence local or transient manifestation does not rescue persistent admissibility. □

The importance of these propositions is methodological. They clarify that later burden arguments need not prove absolute impossibility of local structure. They need only show that the persistence horizon required by the candidate regime outstrips the support horizon available to it.

Corollary 15 (Transient Coherence Corollary)**.** *If the coherence relevant to a regime is real and measurable but supportable only over a finite lifetime or local scale-window, then the regime may appear as a transient precursor or restructuring state without qualifying as a persistent admissible branch.*

Proof. This follows directly from the finite-lifetime obstruction. Local support suffices for transient appearance, but not for the persistence required by stable regime status. □

Corollary 16 (Residual Observability Corollary)**.** *When conventional summarising observables discard or smear the organisation relevant to a regime, coherence-sensitive observables may remain the only structurally meaningful channel through which admissibility-relevant organisation can be detected.*

Proof. If coherence constitutes the organising variable of interest, then observables that are insensitive to coherence may fail even where structure remains present. The correct observable channel must therefore be one that preserves the relevant organisation rather than averaging it away. □

6.2 What the schema does and does not claim

The theorem-schema does not claim that all domains share one metric, one topology, or one operator family. It claims something narrower and more structural: that many domain papers in the corpus can be read as answering the same sequence of questions. What must persist? What coherence does persistence require? What support carries that coherence? How long must it be sustained? What constraints erode it? Does the resulting burden remain below or exceed the admissible budget?

Those questions are transportable even where the technical details are not.

7 Support, Lifetime, and Budget

The theorem-schema becomes clearer when the burden question is decomposed into its three main variables: support, lifetime, and budget. These are not redundant concepts. Each adds something necessary to the regime calculus.

7.1 Support as structural carrying capacity

Support is the effective ability of a system to carry the coherence required by a regime without loss of admissibility. It is never merely the presence of a mechanism. A system may contain phase coupling without supporting persistent phase order, contain nonlinear transport without supporting singular transport, or contain reusable structure without supporting compression at the robustness level required by a forbidden complexity collapse. Support must therefore always be interpreted at the level of carrying capacity.

This distinction matters because many regimes fail not through total absence of support but through insufficiency of support. A system may carry weak, local, or short-lived coherence while

failing to carry global, persistent, or repeatedly renewed coherence. The burden logic of the corpus depends on this distinction. It is not necessary to deny all organisation in order to deny the admissibility of a particular persistent branch.

7.2 Lifetime as admissible persistence horizon

Support becomes meaningful only relative to a persistence horizon. Lifetime names that horizon. It answers the question: for how long, or across what depth, must the coherence remain supportable for the candidate regime to count as a genuine persistent branch?

This notion of lifetime should be read broadly. In empirical systems it may correspond to time before failure, rupture, or release. In transport problems it may correspond to cascade depth or cross-scale renewal length. In analytic or combinatorial domains it may correspond to the extent over which alignment or reuse must remain robust. The unifying point is that persistence without a horizon is too vague to test. Lifetime makes the regime burden operational.

7.3 Budget as a structural ceiling

Budget is the total admissible support available to a regime. It is the ceiling beyond which coherence can no longer be renewed without contradiction, fragmentation, or structural loss. Budget is therefore what turns burden into a real filter rather than a descriptive flourish.

There are several ways a budget may become finite:

- support may decay with scale or persistence depth;
- transport organisation may become unstable under repeated renewal;
- coordination width may be bounded;
- noise, dispersion, or non-commutativity may erode coherence faster than it can be replenished;
- semantic reuse may become fragile under generic or adversarial variation;
- closure itself may fail beyond a certain persistence burden.

The exact mechanism is domain-specific. The function of budget is not.

7.4 Distinct failure modes

The language of budget becomes more useful if one distinguishes several support failure modes.

Decay. Coherence may weaken gradually with scale, depth, or duration. The regime then fails not through sudden rupture but through long-run erosion.

Fragmentation. Coherence may remain present locally while ceasing to integrate into a single regime-level organisation. This is especially important in systems where small coherent patches do not amount to large-scale persistent coherence.

Instability. The support needed to renew coherence may exist in principle but become dynamically unstable under repeated use. Here the issue is not absence but non-robustness.

Summable exhaustion. The support contributions available across layers or scales may be nonzero but summably bounded. In that case finite total carrying capacity still blocks indefinite persistence.

Observational erasure. The regime may remain structurally present in a coherence-sensitive sense while becoming invisible to conventional summary statistics. This is the failure mode relevant to the residual observability branch.

These distinctions matter because they prevent burden from becoming a one-note concept. Different domain papers may instantiate different failure modes while still belonging to the same support-lifetime-budget architecture.

7.5 Local support versus persistent support

A central theme of the paper is the distinction between local support and persistent support. Local support means the system can sustain the relevant organisation within a short horizon, small window, or limited neighbourhood. Persistent support means it can do so across the horizon required for the candidate regime to count as globally or theorem-level significant. Much confusion disappears once this distinction is made explicit.

For instance, the presence of local phase coherence before rupture does not imply the system can carry persistent phase coherence indefinitely. Likewise, the existence of structured alignment at some scales in a mathematical setting does not imply that the alignment can remain coherent across the full horizon needed by the forbidden branch. The burden principle therefore does not deny local structure. It asks whether local support can be promoted to persistent admissibility. Often it cannot.

7.6 Why summability matters

Summability is one of the most useful abstract forms of finite budget. A system may permit repeated contributions to coherence across scales or layers, but if those contributions remain summably bounded, then total carrying capacity remains finite. This provides a powerful way of understanding why nonzero support need not imply admissible persistence.

Summability is therefore not merely a technical convenience. It is a structural bridge between local renewal and global impossibility. It allows a burden argument to concede local organisation while still concluding that the total support available to sustain a persistent pathological regime is insufficient.

7.7 Interim conclusion

Support, lifetime, and budget together form the practical core of the Coherence Burden Principle. Support names what carries organisation. Lifetime names how long that carrying must continue. Budget names the total admissible ceiling. Once these variables are made explicit, the regime logic of the corpus becomes far more transparent. One can then ask, in every case, not merely whether coherence exists, but whether its required burden can be carried for long enough, and at sufficient scale, to justify persistent regime status.

8 Cross-Domain Instantiations

The primary value of the present paper lies in showing that a wide range of existing Darke Theory works can be organised by one regime logic without pretending that they share one technical proof language. The purpose of this section is therefore not to re-prove each paper, but to show how each sits within the burden architecture developed above.

8.1 Riemann Hypothesis

In the Riemann programme, the pathological branch consists of zeros away from the critical line. The argument is not framed as brute-force theorem construction. It is framed as a

burden problem. Off-line persistence would require structured prime-phase coherence or analytic support of a kind that remains organised across the relevant horizon. The burden question is then whether such coherence can remain sustained without collapse, cancellation, or summably bounded depletion. In the language of the present paper, the candidate regime is a persistent pathological regime, the relevant coherence is prime-phase alignment, and the central claim is that its required support burden is not admissibly persistent.

The importance of this reading is that it shifts attention from isolated local structure to persistent obstruction. Even if some local or short-range alignment exists, the regime-level question remains whether the system can support that organisation strongly enough, and for long enough, to house a persistent off-line branch.

8.2 Yang–Mills

In the Yang–Mills paper, the burden variable is not prime-phase organisation but persistent large-scale gauge coherence or holonomy support. Again, the point is not that all gauge coherence is illusory. It is that a particular exceptional regime would require a coherence structure whose support demand outstrips what four-dimensional non-abelian transport can stably carry.

This fits the persistent pathological regime class directly. The hostile-review target is clear: does the forbidden branch really require the large-scale coherence burden claimed, and if so, is the support horizon genuinely finite or unstable in the required sense? The present paper does not answer those domain questions in detail. It shows that the logic of the paper belongs squarely inside the burden architecture.

8.3 Navier–Stokes

The Navier–Stokes work is perhaps the clearest burden-architecture case in the mathematical branch of the corpus. The pathological branch is sustained finite-time singular transport. The burden is expressed through organised cascade or phase-coherent transport across modes and scales. The central structural claim is that the coherence lifetime needed for blow-up does not remain admissibly supportable across the relevant cascade depth.

This case is especially valuable because support, lifetime, and budget are all visible in near-explicit form. The system may support local transfer and bursts of organisation, but that is not enough. The regime significance of singular blow-up depends on a deeper persistence horizon, and the claim is that the necessary burden exceeds what admissible transport organisation can sustain.

8.4 P versus NP

The complexity-theoretic case uses a different technical language, but the structure is recognisably parallel. The relevant coherence is no longer phase or transport coherence, but robust semantic reuse sufficient to compress verification structure while remaining stable under generic or adversarial instance variation. If the required reuse is non-robust or structurally fragile, the persistent compressibility required by the forbidden branch becomes inadmissible.

This case is important because it shows that burden need not be tied to physics or differential equations. The transportable concept is structural carrying capacity, not physical energy. In this setting, the support question becomes: can the system sustain the reuse burden required to collapse verification and search into one admissible robust regime? The burden reading says no.

8.5 Self-organised criticality

The self-organised criticality precursor paper exhibits the positive empirical branch of the present framework. Transient phase coherence appears before large avalanches, exceeds null expectations, collapses during and after failure, and is absent outside transition windows. This does not

describe a forbidden regime. It describes an *admissible transient precursor regime*. The local coherence is real. The long-run persistence is not.

This case is especially important because it prevents the burden framework from being misread as purely negative. Here coherence is not an impossibility condition. It is the signal of a system approaching reconfiguration under finite support.

8.6 Human EEG

The EEG paper extends the same logic into a biological system. Pre-ictal phase organisation substantially exceeds conservative null expectations even while signal power decreases. The crucial point is that structure is carried in coherence rather than in gross amplitude. The present framework classifies this as a transient precursor regime with an additional residual-observability feature, since coherence-sensitive measures retain diagnostic structure under conditions where power summaries do not.

This is one of the strongest arguments for keeping the residual-observability class distinct. If one looked only at amplitude or power, much of the relevant organisation would be missed. Coherence here functions both as precursor burden and as observational access route.

8.7 Seismology and volcanic destabilisation

The Kaikōura and Whakaari papers fit naturally into the same regime logic. Failure is not treated as a purely local energetic spike. Instead, rupture and destabilisation are preceded by reorganised transport or phase structure. The support is real but finite-lived. These papers therefore reinforce the precursor branch rather than the exclusion branch.

They also show why lifetime matters. The relevant structure appears not as a permanent background order, but as a pre-event organisation that sharpens as the system approaches transition. The burden perspective therefore interprets these findings as finite support manifestations rather than contradictions of the general principle.

8.8 Turbulence and solar plasma

The turbulence and solar-corona studies likewise point to phase-structured transport before macroscopic release or strong nonlinear organisation. In the present language, they support the claim that coherence often appears most clearly as a local pre-release burden rather than as a permanently admissible global branch.

These cases are particularly helpful because they sit close to the transport and cascade language used in the mathematical exclusion branch, especially in Navier–Stokes. They therefore form a natural bridge between mathematical burden arguments and empirical precursor detection.

8.9 Radar and sensing

The radar paper is especially useful because it clarifies a third role for coherence. In cluttered, low-SNR settings, energy-only summaries may fail even when coherence-sensitive observables still retain structure. This makes radar a canonical example of a structurally informative residual regime: the important organisation survives, but only in the correct observational channel.

This branch matters for the future of the corpus because it implies that burden-based structure may remain detectable even where conventional descriptive channels fail. In that sense observability is not an afterthought to admissibility. It is part of how one learns what the real organisation of the system is.

8.10 Cosmology and astrophysical structure

The cosmological papers suggest that the same regime logic may operate at still larger scales. Inflation is recast through refinement pressure and closure burden rather than through independently imposed inflationary machinery. Directional and void-conditioned variation in the local Hubble relation is read through conditioned closure geometry rather than homogeneous-background assumptions alone. Little Red Dots are interpreted as early closure-dominated structures rather than as anomalies to be absorbed by conventional bookkeeping.

These works are not all proofs of the same type, and this paper does not pretend otherwise. Their unity is structural rather than technical. Each asks, in one form or another, what large-scale branches can actually be supported by bounded refinement and closure-bearing persistence. That is exactly the sort of question the burden principle is designed to organise.

8.11 Why this section matters

The purpose of the cross-domain section is not to claim that all later papers are equally strong or equally formal. It is to show that they increasingly occupy related positions within a common regime architecture. Once this is recognised, the corpus looks less like an expanding list of applications and more like a theory that is discovering its own canonical variables.

9 Hostile Review Criteria

The current paper is only as strong as the pressure points it exposes. A serious hostile review should not attack the work at the level of tone, analogy, or broad philosophical disagreement alone. It should attack the exact transitions on which the burden logic depends.

1. Whether the pathological or precursor regime truly requires the claimed coherence.
2. Whether the notion of coherence is made precise enough in the relevant domain.
3. Whether the support budget is genuinely finite, summable, unstable, or otherwise bounded in the way claimed.
4. Whether the candidate regime could survive through a weaker, more fragmented, or more localised structure than the paper permits.
5. Whether the lifetime required by the regime is correctly identified.
6. Whether the relevant observables genuinely track structure rather than artefacts of amplitude, variance, windowing, or preprocessing.
7. Whether there exist countermodels in which the burden is paid differently from what the argument assumes.
8. Whether the classification into persistent, transient, or residual regime types has been applied correctly.

These are the proper review criteria because they target the exact logical hinge of the programme:

$$\begin{aligned} \text{persistent pathology} &\Longrightarrow \text{required coherence} \\ &\Longrightarrow \text{support burden over lifetime} \\ &\Longrightarrow \text{inadmissibility or finite-lived manifestation.} \end{aligned}$$

If that chain fails, the argument fails. If the chain holds, the candidate regime is structurally constrained in exactly the way the burden principle predicts.

The value of listing these criteria openly is not merely defensive. It is disciplinary. A programme matures by becoming more vulnerable at the correct points. The present paper therefore aims to make future burden-based arguments easier to attack properly, not harder.

10 Limits of the Principle

The present framework inherits the internal limits already established for Darke Theory more broadly. These limits should be stated explicitly, because the power of the principle depends upon its refusal to overclaim.

1. It is not a theory of dynamical law generation. It does not derive equations of motion from first principles.
2. It does not decide between multiple regimes that are each structurally admissible. Admissibility is a filter, not a complete selector.
3. It does not apply where no coherent notion of burden, support, or lifetime can be defined.
4. It does not replace the technical labour required in any given domain. The burden schema organises domain arguments; it does not discharge them automatically.
5. It does not convert empirical suggestiveness into proof. Transient coherence evidence remains evidence, not theorem.
6. It does not authorise metaphorical extension into domains where coherence and constraint cannot be made structurally precise.

These are not weaknesses to be hidden. They are part of the paper's discipline. The principle is powerful only because it is selective. Where coherence, persistence, closure, and contradiction-inducing forbidden regimes cannot be stated with clarity, the framework should refuse speculation.

11 Toward a General Regime Theory

The main conceptual contribution of the paper may now be stated plainly. The Darke Theory corpus is not merely a collection of foundations, applications, and proof attempts. It is converging on a general regime theory in which persistent structure, precursor structure, and residual observability are all organised by the same deeper question:

> What coherence must a regime carry, for how long, and under what support constraints?

Under this view, the corpus divides naturally into three branches.

- The *exclusion branch*, where persistent pathology is ruled out because its required coherence burden exceeds admissible support.
- The *precursor branch*, where coherence appears as finite-lived organisation before reconfiguration, rupture, or failure.
- The *observability branch*, where coherence-sensitive measurements preserve structure that ordinary summaries discard.

This is a stronger organising picture than one obtains by speaking only of inadmissibility in the abstract. It shows not merely that the later papers share a style, but that they occupy related places within a common regime architecture. It also suggests that support, lifetime, and burden may prove to be the canonical regime variables by which Darke Theory classifies an increasing share of its future work.

12 Conclusion

This paper has proposed the Coherence Burden Principle as a transportable law already implicit across a substantial portion of the Darke Theory corpus. The principle states that a candidate regime is structurally admissible only if the coherence required for its persistence can be supported for the necessary lifetime within the closure, refinement, and constraint structure of the governing system.

From this principle, three classes of regime become visible: persistent pathological regimes, which fail because their burden exceeds admissible support; transient precursor regimes, which appear locally and meaningfully but cannot persist as stable branches; and structurally informative residual regimes, in which coherence-sensitive observables remain useful after conventional summaries collapse.

The result is not a replacement for the domain papers. It is a bridge among them. It explains how mathematical exclusion, empirical precursor structure, and coherence-based sensing can all belong to one methodological lineage without sharing one technical proof language. In that lineage, coherence is the currency of persistence, constraint is the mechanism of its erosion, and admissibility is the filter that decides what can be stably borne.

If the wider corpus continues in this direction, then support, lifetime, and burden may prove to be not merely recurring motifs, but the canonical variables by which Darke Theory classifies regimes across domains.

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